插图英语阅读小册

第6册 · 十篇故事

按推荐顺序阅读，保留原篇号。点击篇目即可跳转；每篇依次包含插图、英文原文、中文译文、词汇与回想。

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林下之网

The Web Beneath the Woods



树木的地下联系，哪些已有证据?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

One of ecology's most romantic ideas grew out of plastic bags. In a Canadian forest, researchers enclosed paper birch and Douglas fir seedlings in them, then bathed each species in carbon dioxide laced with a different tracer. Carbon shuttled both ways, apparently via fungi linking the trees' roots, yet the firs ran a slight surplus, larger in deeper shade.

Most terrestrial plants harbour such fungi, whose filaments forage for water and nutrients including nitrogen, swapping them for plant sugars. One fungus can knit plants into a shared network. The journal Nature unveiled the study on 7 August 1997, its cover offering a catchy coinage: the "wood-wide web".

The metaphor proved irresistible. Between 2015 and 2021, a forester's bestseller, an acclaimed novel, and an ecologist's memoir told readers that trees converse, warn one another, and, as "mother trees", nurture their kin. Yet the 1997 trees were seedlings, not elderly mothers.

In February 2023, three researchers combed through the field evidence. They judged it too sparse, inconsistent, or ambiguous to demonstrate that shared networks are widespread. In 26 field studies, networks helped and hindered seedlings about equally often, with neutral effects predominating. No peer-reviewed field study showed mature trees favouring their own offspring. Unsupported assertions in later papers had doubled over 25 years, with citations skewed towards positive results.

None of this disproves the networks' existence. In January 2025, defenders faulted the critique's methods. The critics concede that carbon and nutrients can migrate between trees underground. It is evidence unearthed in real forests, not the allure of a family saga, that must settle how much the networks convey, and whether it matters.

中文译文

生态学中一个格外浪漫的观念，竟起源于塑料袋。在加拿大的一片森林里，研究人员把纸桦和花旗松的幼苗罩进袋中，分别让两种幼苗接触掺有不同示踪物的二氧化碳。碳在两种树之间双向转移，似乎经由连接根系的真菌传递。不过，花旗松最终获得了少量净盈余，在遮荫更深的地方，盈余也更大。

大多数陆生植物都与这类真菌共存。真菌的菌丝寻找水分和包括氮在内的养分，再拿这些资源换取植物的糖分。一个真菌就能将多株植物连成共享网络。《自然》于1997年8月7日发表这项研究，封面还用上了一个好记的新词：“林木万维网”。

这个比喻实在太有吸引力。2015年至2021年间，一位护林人的畅销书、一部广受好评的小说和一位生态学家的回忆录，都告诉读者：树木能够交谈、相互示警，还会像“母树”一样养育亲族。可1997年实验中的树只是幼苗，并不是年迈的母亲。

2023年2月，三位研究者系统梳理了野外证据。他们认为，这些证据或太少、或互不一致、或含糊不清，不足以证明共享网络普遍存在。在26项野外研究中，帮助与妨碍幼苗的结果出现得大致同样频繁，而最多见的是没有明显影响。没有一项经同行评审的野外研究表明，成年树木会偏爱自己的后代。后续论文中缺乏依据的断言在25年间翻了一倍，引用也偏向正面结果。

这些批评并不能否定网络的存在。2025年1月，网络说的拥护者质疑了那篇批评文章的方法；批评者也承认，碳和养分能够在树木之间经地下迁移。网络究竟传输了多少、这种传输在生态上有多大作用，必须由真实森林中的证据来判定，不能取决于一段“树木家族传奇”有多动人。

“无明显影响最多”与“帮助、妨碍的次数接近”并不矛盾；复核未证明网络不存在。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ecology n.
生态学
One of ecology's most romantic ideas

seedling n.
幼苗
paper birch and Douglas fir seedlings

tracer n.
示踪物
a different tracer

apparently adv.
似乎；看来
apparently via fungi

surplus n.
盈余
a slight surplus

terrestrial adj.
陆生的
Most terrestrial plants

filament n.
丝状体；此处指菌丝
whose filaments forage for water

nutrient n.
养分
water and nutrients including nitrogen

metaphor n.
比喻
The metaphor proved irresistible

nurture v.
养育；培育
nurture their kin

comb through 动词短语
仔细梳理；彻底搜查
combed through the field evidence

sparse adj.
稀少的
too sparse, inconsistent, or ambiguous

inconsistent adj.
互不一致的
inconsistent, or ambiguous

ambiguous adj.
含糊不清的
inconsistent, or ambiguous

demonstrate v.
证明
to demonstrate that shared networks

widespread adj.
普遍存在的
shared networks are widespread

neutral adj.
无明显正负影响的
neutral effects predominating

peer-reviewed adj.
经同行评审的
No peer-reviewed field study

assertion n.
断言
Unsupported assertions

citation n.
引文；引用
with citations skewed

concede v.
承认（先前不愿承认的事）
The critics concede

convey v.
传输；输送
how much the networks convey

文中词汇与搭配

补充词汇共50项。保留每个表达在本文中的意思。

romantic adj.
富有浪漫色彩的

grow out of 动词短语
源自；由……发展而来

enclose v.
把……围住；罩住

in prep.
在……里面

bathe v.
使浸在；使充分接触

be laced with 动词短语
掺有

shuttle v.
往返移动

fungus n.
真菌

link v.
连接

slight adj.
少量的；轻微的

shade n.
遮荫

harbour v.
容纳；携带

forage for 动词短语
搜寻（水或养分）

nitrogen n.
氮

swap v.
交换

knit v.
联结；编织

into prep.
进入；连成

shared network n.
共享网络

unveil v.
公布；发表

catchy adj.
好记而吸引人的

coinage n.
新造词

irresistible adj.
难以抗拒的

bestseller n.
畅销书

acclaimed adj.
广受赞誉的

memoir n.
回忆录

文中词汇与搭配 续

补充词汇共50项。保留每个表达在本文中的意思。

converse v.
交谈

warn one another 动词短语
互相警告

kin n.
亲族

elderly adj.
年迈的

hinder v.
妨碍

predominate v.
占多数

mature adj.
成熟的；成年的

favour v.
偏爱

offspring n.
后代

unsupported adj.
缺乏依据的

double v.
翻倍

skew v.
使偏向

towards prep.
朝；偏向

positive result n.
正面结果

disprove v.
证明……不成立

existence n.
存在

defender n.
拥护者

fault v.
指责；挑出问题

critique n.
批评性分析

critic n.
批评者

migrate v.
迁移

unearth v.
发掘

allure n.
吸引力

saga n.
传奇故事

matter v.
有重要意义

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Carbon shuttled both ways, apparently via fungi linking the trees' roots, yet the firs ran a slight _____, larger in deeper shade.

02 The _____ proved irresistible.

03 In 26 field studies, networks helped and hindered seedlings about equally often, with _____ effects predominating.

04 The critics _____ that carbon and nutrients can migrate between trees underground.

文章主线

- 发现 1997年，示踪实验发现纸桦和花旗松幼苗双向交换碳，似乎经连接根系的真菌传递。
- 走红 真菌以水和养分换植物糖分，也能连接多株植物；“林木万维网”走红后，被演绎成母树哺育亲族。
- 复核 2023年复核认为网络普遍存在的证据不足；26项野外研究以无明显影响为主，帮助和阻碍幼苗的次数相近。
- 偏差 复核未找到母树偏爱亲生后代的同行评审野外证据；论文偏引正面结果，无依据断言在25年间翻倍。
- 争议 2025年拥护者质疑复核方法；批评者也承认树间有地下物质转移，但网络输送多少、生态作用多大仍待实证。

参考答案　1 surplus 2 metaphor 3 neutral 4 concede

向鸸鹋宣战

The Great Emu War



机枪为何没能解决鸸鹋危机?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Around Campion, Western Australia, most farmers were First World War veterans, settlers placed by a government scheme on marginal parcels of farmland. They were inexperienced growers, and a quarter had already quit. In 1932, after drought, roughly 20,000 emus swarmed in from arid country in flocks and trampled the wheat crops.

The farmers' plight took a military turn. In desperation, a delegation of ex-soldiers aired their grievances and asked the Defence Minister for machine-guns. He agreed, sending not a battalion but a detachment: Major Meredith, an artillery officer, with two soldiers, two Lewis guns, and 10,000 rounds of ammunition. The offensive began on 2 November 1932.

At a dam, about a thousand birds approached an ambush, but a gun jammed after a dozen fell. The flocks broke formation into small, agile groups, frustrating each tactical plan, and a truck-mounted gun proved futile. Six days of wasteful firing spent 2,500 rounds on perhaps 50 to 200 birds, a ludicrous return. After parliamentary questions and press ridicule, an embarrassing withdrawal came on 8 November, though shooting restarted within days.

Meredith was recalled on 10 December. His memo's inventory counted 986 kills from 9,860 rounds, a ratio of ten bullets to one bird. He likened the resilient birds' toughness to the armour of tanks. Five weeks of warfare had barely dented the hardy flocks' endurance.

Later pleas for reinforcements were refused, and intervention gave way to a bounty sanctioned by the government. This payout per bird gave farmers compensation and an incentive for extermination. Farmers claimed over 57,000 payments in six months of 1934, more than fifty-seven times Meredith's 986 kills.

中文译文

在西澳大利亚州坎皮恩一带，多数农民是第一次世界大战的退伍军人；政府通过一项安置计划，让这些定居者在一块块贫瘠农地上定居。他们缺少种植经验，已有四分之一放弃了耕作。1932年旱灾过后，约两万只鸸鹋从干旱地区成群涌入，践踏小麦庄稼。

农民的困境由此走向了军事解决。在绝望之中，一支由退伍士兵组成的代表团公开诉说不满，向国防部长请求机枪。部长答应了；派来的却不是一个营，而是一支小分队：炮兵军官梅雷迪思少校、两名士兵、两挺刘易斯机枪，以及一万发弹药。行动于1932年11月2日开始。

在一座水坝旁，约一千只鸸鹋接近伏击地点；但只打倒十来只，一挺机枪便卡了壳。鸟群散成灵活的小群，使一个个战术计划落空，卡车上架设的机枪也未能奏效。六天中低效的射击消耗了2500发子弹，可能只打死50至200只鸟，成效实在令人啼笑皆非。议会提出质询，报界加以嘲讽，部队于11月8日尴尬撤退，不过几天后便又恢复了射击。

梅雷迪思于12月10日被召回。他的备忘录列出的清单显示，9860发子弹打死986只鸟，相当于每只鸟耗费十发子弹。他把这些鸟顽强的抗击打能力比作坦克装甲。五个星期的战事，几乎没能削弱这些顽强鸟群的耐力。

此后，请求增援的恳求都遭到拒绝，军事干预让位于政府批准的悬赏制度。按每只鸟发放的赏金，既给农民提供了补偿，也成为扑杀鸸鹋的激励。1934年的六个月内，农民申领了五万七千多笔赏金，数量是梅雷迪思所报986只击杀数的五十七倍多。

核心词汇

先记这22项。连着下方的原文短语一起回想。

veteran n.
退伍军人
First World War veterans

marginal adj.
贫瘠的；不适宜耕作的
on marginal parcels of farmland.

arid adj.
干旱的
from arid country in flocks

plight n.
困境；苦境
farmers' plight

delegation n.
代表团
a delegation of ex-soldiers

grievance n.
不满；委屈
their grievances

detachment n.
小分队；分遣队
not a battalion but a detachment

ammunition n.
弹药
10,000 rounds of ammunition.

ambush n.
伏击；埋伏
approached an ambush

agile adj.
灵活的；敏捷的
small, agile groups

frustrate v.
使受挫；使落空
frustrating each tactical plan

futile adj.
徒劳的；无效的
proved futile.

ludicrous adj.
荒唐可笑的
a ludicrous return.

withdrawal n.
撤退；撤离
an embarrassing withdrawal

ratio n.
比例；比率
a ratio of ten bullets to one bird.

resilient adj.
顽强的；承受力强的
the resilient birds' toughness

reinforcement n.
增援部队；援兵
pleas for reinforcements

intervention n.
干预；介入
intervention gave way to a bounty

give way to phr. v.
让位于；被……取代
intervention gave way to a bounty

bounty n.
赏金；悬赏
a bounty sanctioned by the government.

compensation n.
补偿；赔偿
gave farmers compensation

incentive n.
激励；诱因
an incentive for extermination.

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

settler n.
定居者；移民

parcel n.
一块（土地）

farmland n.
农地

inexperienced adj.
缺乏经验的

grower n.
种植者

swarm v.
蜂拥而至

in flocks phr.
成群地（尤指鸟）

trample v.
践踏；踩坏

crop n.
庄稼；农作物

in desperation phr.
在绝望之中

air v.
公开表达

machine-gun n.
机枪

battalion n.
营（军队编制）

artillery n.
炮兵；火炮

offensive n.
进攻；攻势

jam v.
卡住；卡壳

formation n.
队形

tactical adj.
战术的

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

wasteful adj.
浪费的

parliamentary adj.
议会的

ridicule n.
嘲笑；嘲讽

embarrassing adj.
令人尴尬的

memo n.
备忘录

inventory n.
清单

bullet n.
子弹

liken v.
把……比作

toughness n.
坚韧；耐打程度

to prep.
与liken连用，表示比作

armour n.
装甲

warfare n.
战事；战争

hardy adj.
顽强的；耐受力强的

endurance n.
耐力；持久力

plea n.
恳求

sanction v.
批准；准许

payout n.
支付的款项

extermination n.
彻底消灭；扑灭

payment n.
支付款项

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The farmers' _____ took a military turn.

02 After parliamentary questions and press ridicule, an embarrassing _____ came on 8 November, though shooting restarted within days.

03 His memo's inventory counted 986 kills from 9,860 rounds, a _____ of ten bullets to one bird.

04 Later pleas for _____ were refused, and intervention gave way to a bounty sanctioned by the government.

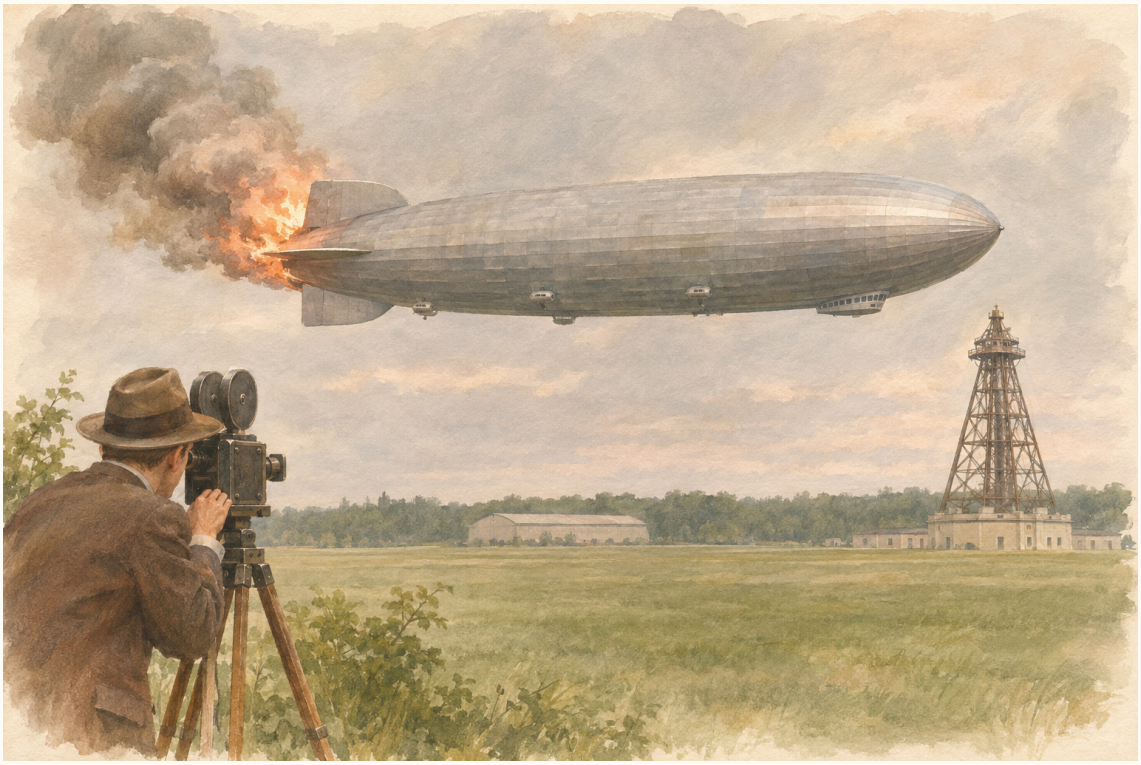
文章主线

- 困境 1932年旱灾后，约两万只鸸鹋涌入西澳农田、践踏庄稼，当地退伍军人农户向军方求助。
- 出兵 军方派一名少校和两名士兵，携两挺机枪、一万发子弹，试图控制鸟群。
- 受挫 鸟群分散成灵活小群，机枪卡壳，车载射击也无效，行动受到媒体和议会嘲讽。
- 撤军 部队一度撤离又复战，五周后被召回；报告称9860发子弹击杀986只，仍未解决鸟患。
- 转向 政府拒绝增援，改按只发赏金；1934年半年内农民申领逾5.7万笔，远超军方战果。

参考答案　1 plight 2 withdrawal 3 ratio 4 reinforcements

兴登堡号的 34 秒

The Last Flight of the Hindenburg



一场被拍下的大火，怎样终结了飞艇客运时代？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1936 the colossal Hindenburg made 17 Atlantic round trips, a marvel of luxury and glamour. Its amenities included a lounge and, that year, a grand piano. An American embargo on helium export had left the ship relying on hydrogen. Because that gas was combustible, smokers sat behind an airlock with one electric lighter.

The airship left Frankfurt on 3 May 1937. Hours behind its timetable, it met thunderstorms at Lakehurst and circled for over an hour before its descent towards the mooring mast. At about 7:25 pm on 6 May, flames appeared near the stern. The inferno incinerated the ship in about half a minute, leaving a crumpled frame.

Of the 97 aboard, 62 survived, a proportion close to two-thirds, while a minority of 35 died, as did one man on the ground. A 14-year-old cabin boy was drenched by a deluge from a burst water tank, and the soaking saved him. He dropped through a hatch and scrambled clear, escaping only narrowly.

Newsreel cameras and photographers caught the disaster. A radio correspondent's eyewitness commentary was not live; audiences heard it the next day. The cause has not been definitively established, but static electricity igniting leaking hydrogen is the leading explanation. A theory blaming the doped fabric has been rejected.

The 1933 crash of USS Akron killed 73 yet won less notoriety. The filmed fire left a fateful image that abruptly wrecked confidence. As public perception soured, Graf Zeppelin was withdrawn, later scrapped, and a glamorous epoch of aviation closed. The doomed airship's first journey to North America had begun on 6 May 1936, exactly a year before the fire.

中文译文

1936年，庞大的兴登堡号完成17次跨大西洋往返，堪称奢华而迷人的奇观。艇内设有休息厅，那一年还配过一架三角钢琴。由于美国禁止出口氦气，飞艇只能依靠氢气升空。氢气易燃，吸烟者只能待在气闸隔开的区域，共用一个电点烟器。

1937年5月3日，飞艇离开法兰克福。它抵达莱克赫斯特时已晚点数小时，又遇上雷暴，盘旋一个多小时后才开始向系留塔下降。5月6日晚约7点25分，尾部附近出现火焰。大火在约半分钟内焚毁飞艇，只留下一副扭曲的骨架。

艇上97人中，62人生还，接近三分之二；35人遇难，地面另有一名男子死亡。一名14岁的艇上服务生被破裂水箱涌出的水浇透，这一身湿水救了他。他从舱口落下，连滚带爬逃到安全处，险些未能脱身。

新闻纪录片摄像机和摄影师拍下了灾难。电台记者的现场目击解说并非直播，听众到第二天才听到。起火原因至今没有定论，主要解释是静电点燃泄漏的氢气。把起火归咎于蒙皮织物涂层的说法已被否定。

1933年阿克伦号失事造成73人死亡，却没有那么广为人知。被拍下的大火留下了一幅影响深远的画面，骤然摧毁公众信心。随着公众看法恶化，格拉夫·齐柏林号退出运营，后来被拆解，航空史上一个充满魅力的时代就此落幕。兴登堡号首次飞往北美的航程始于1936年5月6日，恰好是失火前整整一年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

colossal adj.
庞大的
the colossal Hindenburg

amenity n.
便利舒适设施
Its amenities included a lounge

embargo n.
禁运
An American embargo on helium export

combustible adj.
易燃的
that gas was combustible

descent n.
下降
its descent towards the mooring mast

mooring mast 名词短语
飞艇系留塔
towards the mooring mast

stern n.
飞艇尾部
near the stern

inferno n.
猛烈的大火
The inferno incinerated the ship

incinerate v.
焚毁
The inferno incinerated the ship

crumple v.
压皱；扭曲变形
leaving a crumpled frame

proportion n.
比例
a proportion close to two-thirds

drench v.
浇透；使湿透
was drenched by a deluge

deluge n.
大量涌出的水
a deluge from a burst water tank

hatch n.
舱口
He dropped through a hatch

correspondent n.
新闻记者
A radio correspondent's

eyewitness adj.
亲眼目击的
eyewitness commentary was not live

commentary n.
现场解说
eyewitness commentary was not live

definitively adv.
确切地；有定论地
has not been definitively established

static electricity 名词短语
静电
static electricity igniting

leading explanation 名词短语
主要解释
is the leading explanation

perception n.
看法；观感
As public perception soured

epoch n.
时代
a glamorous epoch of aviation

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

round trip 名词短语
往返航程

marvel n.
令人惊叹的事物

luxury n.
奢华

glamour n.
魅力

lounge n.
休息厅

helium n.
氦气

export n.
出口

hydrogen n.
氢气

smoker n.
吸烟者

sit behind 动词短语
坐在……隔开的后方

electric lighter 名词短语
电点烟器

airship n.
飞艇

timetable n.
时刻表

thunderstorm n.
雷暴

circle v.
盘旋

before prep.
在……之前

flame n.
火焰

frame n.
骨架；框架

two-thirds 数词
三分之二

a minority of 名词短语
少于半数的

cabin boy 名词短语
艇上少年服务生

burst v.
爆裂

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

soaking n.
全身湿透

scramble v.
急忙攀爬；连滚带爬

narrowly adv.
勉强；险些未能

photographer n.
摄影师

audience n.
听众

ignite v.
点燃

leak v.
泄漏

theory n.
解释；假说

doped adj.
涂有涂料的

fabric n.
蒙皮织物

reject v.
否定；不接受

notoriety n.
因坏事而广为人知

fateful adj.
影响深远而不幸的

image n.
画面

abruptly adv.
突然地

confidence n.
信心

sour v.
变坏；恶化

withdraw v.
撤出运营

scrap v.
报废拆解

glamorous adj.
富有魅力的

aviation n.
航空

doomed adj.
注定毁灭的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 An American _____ on helium export had left the ship relying on hydrogen.

02 Because that gas was _____,

03 The cause has not been _____ established,

04 As public _____ soured,

文章主线

- 隐患 兴登堡号是豪华跨洋飞艇；美国禁运氦气，使它只能靠易燃氢气浮空，留下安全隐患。
- 失事 1937年5月6日，它在美国准备停靠系留塔时尾部起火，约半分钟就烧成扭曲骨架。
- 伤亡 艇上97人中62人生还、35人遇难；地面另有1人死亡，遇难者合计36人。
- 原因 火因尚无定论，主要解释是静电点燃泄漏氢气；归咎外层布料涂层的说法被否定。
- 落幕 1933年另一飞艇事故死73人，却不及兴登堡号出名；这次影像和次日播出的录音重创信心，飞艇客运时代落幕。

参考答案　1 embargo 2 combustible 3 definitively 4 perception

比信先到家的鸟

The Bird That Beat the Letter Home



能辨方向的鸟，怎样知道自己在哪里、家又在哪里？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 3 June 1952 Rosario Mazzeo released a Manx shearwater in Boston. It came by rail and airliner from its burrow on Skokholm, an isle off Wales, and Mazzeo posted a letter at once. After a roughly 3,200-mile odyssey across the unbroken Atlantic, the feathered traveller was back on its doorstep twelve and a half days later. The letter arrived afterwards.

The first hints of a compass were celestial. In the 1950s Kramer kept starlings cooped in cages and moved the apparent sun with mirrors. The duped birds swivelled to match, allowing for the sun's daily arc. In the 1960s Emlen curtained off stars in a planetarium's vault. His buntings steered by stars around the pole-star, the axis of the nightly whirl.

In 1972 the Wiltschkos showed that robins read Earth's magnetic field by inclination, not polarity, unlike a sailor's needle. In 1978 Schulten proposed a light-triggered reaction to decode the field. Proposed in 2000, cryptochrome, a protein in the retina, is now the foremost suspect. In 2021 robin cryptochrome proved more finely tuned in the laboratory than that of chickens or pigeons. Proof in a living bird is awaited.

In 2014 came an eerie, man-made twist. At Oldenburg University, robins in campus huts lost that sense. In huts clad in grounded aluminium, they reclaimed their bearings. Removing the grounding, or piping radio noise in, bewildered them again in double-blind tests. The interference, a murmur from everyday appliances, sat far below WHO safety limits.

How a wanderer knows where it is remains unanswered. The compass code is partly cracked, but the atlas that steered the shearwater has yet to be read.

中文译文

1952年6月3日，罗萨里奥·马泽奥在波士顿放飞一只大西洋鹱。它原先栖居在威尔士外海斯科克霍姆岛的巢穴中，经火车和客机运到这里；马泽奥放飞后立即寄出一封信。跨越约3200英里连绵无际的大西洋后，这位长着羽毛的旅客于十二天半后回到巢门口。信却更晚才到。

最早的“罗盘”线索来自天体。20世纪50年代，克雷默把椋鸟关在笼中，用镜子改变鸟看到的太阳位置。受骗的鸟随之转向，同时还能把太阳每日运行的轨迹考虑进去。20世纪60年代，埃姆伦在天文馆穹顶上遮挡部分星星；他的鸥鸟依靠北极星周围的星星辨向，那里是夜空看似每夜旋转的中心。

1972年，维尔奇科夫妇证明，知更鸟依靠地磁倾角，而不是磁场极性来感知地磁，这与水手的罗盘指针不同。1978年，舒尔滕提出，它们可能利用光触发的反应解读磁场。2000年，有人提出视网膜中的隐花色素蛋白可能参与其中，如今它已成为头号候选者。2021年，实验室研究显示，知更鸟隐花色素的磁感反应，比鸡或鸽子的更精细；但仍需在活鸟体内证明这一机制。

2014年，事情出现了一个奇异的人为转折。奥尔登堡大学校园小屋中的知更鸟失去了这种定向感；在包覆接地铝板的小屋内，它们却恢复了方向感。双盲实验中，无论断开接地，还是引入无线电噪声，都让鸟再次迷向。干扰来自日常电器产生的微弱电磁噪声，强度远低于世卫组织的安全限值。

鸟儿究竟怎样知道自己身在何处，仍无答案。罗盘的密码已被破解一部分，但指引那只大西洋鹱回家的“地图”，还没有人读懂。

地磁倾角是地磁方向与水平面的夹角。文中的世卫安全限值针对人体暴露，不能据此判断鸟类定向是否受干扰。

核心词汇

先记这22项。连着下方的原文短语一起回想。

odyssey n.
漫长艰辛的旅程
a roughly 3,200-mile odyssey

celestial adj.
天体的；天空的
hints of a compass were celestial

apparent sun 名词短语
鸟所看到的太阳位置
moved the apparent sun with mirrors

allow for 动词短语
考虑到；作出相应调整
allowing for the sun's daily arc

axis n.
轴心
the axis of the nightly whirl

magnetic field 名词短语
磁场
Earth's magnetic field

inclination n.
地磁倾角；地磁方向与水平面的夹角
by inclination, not polarity

polarity n.
磁场的南北极性
by inclination, not polarity

light-triggered adj.
由光触发的
a light-triggered reaction

decode v.
解读；破解
to decode the field

retina n.
视网膜
a protein in the retina

foremost adj.
最主要的；首要的
the foremost suspect

suspect n.
待验证的主要候选对象
the foremost suspect

await v.
等待；有待取得
Proof in a living bird is awaited

reclaim v.
找回；恢复
they reclaimed their bearings

bearings n.
方向感；方位
they reclaimed their bearings

bewilder v.
使迷向；使不知所措
bewildered them again

interference n.
电磁干扰
The interference, a murmur

appliance n.
家用电器
everyday appliances

unanswered adj.
尚无答案的
remains unanswered

crack v.
破解
The compass code is partly cracked

atlas n.
地图集；此处比喻导航地图
the atlas that steered the shearwater

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

release v.
放飞；释放

rail n.
铁路；火车运输

airliner n.
民航客机

burrow n.
地洞；洞穴式巢穴

isle n.
小岛

post a letter 动词短语
寄出一封信

at once 介词短语
立即

unbroken adj.
连续无间的；连绵的

Atlantic 专名
大西洋

feathered adj.
长着羽毛的

traveller n.
旅行者

doorstep n.
门口；此处比喻巢口

afterwards adv.
后来；随后

hint n.
线索；迹象

coop v.
关在狭小空间里

cage n.
笼子

dupe v.
欺骗

swivel v.
转动方向

arc n.
弧线；运行轨迹

curtain off 动词短语
用帘子遮挡

vault n.
穹顶

steer v.
辨向；调整方向

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

pole-star n.
北极星

nightly adj.
每夜的

whirl n.
旋转

robin n.
知更鸟

sailor n.
水手

needle n.
罗盘指针

finely adv.
精细地

tune v.
调节；使敏锐

pigeon n.
鸽子

eerie adj.
奇异得令人不安的

man-made adj.
人为的

twist n.
意外转折

hut n.
小屋

sense n.
感知能力

clad in 分词短语
包覆着

aluminium n.
铝

pipe v.
输送；引入

in adv.
进来；与pipe连用

murmur n.
微弱声响；此处比喻微弱噪声

safety limit 名词短语
安全限值

wanderer n.
漫游者

have yet to 动词短语
尚未

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The first hints of a compass were _____.

02 robins read Earth's magnetic field by _____, not polarity

03 Proof in a living bird is _____.

04 The compass code is partly _____

文章主线

- 归巢 大西洋鹱被运到波士顿放飞，12天半跨越约3200英里回巢，比放飞者随即寄出的信还早。
- 罗盘 用镜子改变鸟看到的太阳位置、在天文馆遮星的实验，分别证实不同鸟类能借太阳和星空辨向。
- 机制 知更鸟按地磁倾角辨向；眼内隐花色素蛋白在实验室显示磁感潜力，可能通过光触发的反应感磁。
- 干扰 双盲实验中，低于世卫限值的电器电磁干扰也会使知更鸟迷向；接地铝板恢复辨向，断开接地又迷失。
- 未解 隐花色素感磁尚待活鸟验证；能辨方向不等于知道自己在哪，鸟如何确定回巢路线仍未解。

参考答案　1 celestial 2 inclination 3 awaited 4 cracked

大恶臭逼出了下水道

The Stink That Built London's Sewers



下水道本为消除臭味，为什么还能减少疾病?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In the fierce heat of summer 1858, the curtains of the new Houses of Parliament hung heavy with chloride of lime, a bleach to make the air tolerable. The Thames festered, and MPs contemplated moving out.

The river had become London's lavatory. Antiquated sewers and outlets discharged the untreated sewage of some three million city dwellers into its currents, where the excrement stewed and decayed. The vile fumes grew so overpowering that the episode earned a name, the Great Stink.

Disgust found a voice on 15 July, when the Chancellor of the Exchequer, Benjamin Disraeli, denounced the river as a hellish pool of unbearable horrors. A bill funding new sewers was fast-tracked into law in about 18 days.

The engineer Joseph Bazalgette masterminded the response. His 82 miles of stout intercepting sewers caught the flow short of the river and diverted it east, downstream of the city. With some 1,100 miles of street sewers as tributaries, the infrastructure took 318 million bricks. Having calculated an adequate calibre for each pipe, he doubled it. The backbone lies buried underneath the newly built Victoria, Albert, and Chelsea Embankments. Substantially complete by 1875, much of the network is in use today.

Yet that sanitation rested on a fallacy. The predominant belief, miasma, blamed disease on foul air and vapours, so the sewers were intended principally to suppress the smell. They worked for an unintended reason, guarding the purity of the water Londoners drank.

Cholera resurfaced in 1866 and killed about 5,600 people in the East End, the one locality not yet connected.

A remedy aimed at the nose protected the gut.

中文译文

1858年夏天酷热难耐。新议会大厦的窗帘浸了含氯漂白粉，沉甸甸地垂着，只为让空气勉强可以忍受。泰晤士河腐败发臭，议员们考虑搬离。

这条河已成伦敦的厕所。老旧下水道和排污口，把约300万市民未经处理的污水排进河中，粪便在暑热中沤积、腐烂。恶臭强烈得令人无法忍受，这场事件因此有了一个名字：“大恶臭”。

7月15日，财政大臣本杰明·迪斯雷利替这份厌恶发了声，公开斥责河流是一潭地狱般、恐怖得让人无法忍受的污水。为新下水道拨款的法案加速审议，约18天便成为法律。

工程师约瑟夫·巴泽尔杰特主持设计这套应对工程。82英里坚固的截流下水道，在污水流入河中之前将其截住，改道向东，送往城市下游。约1100英里的街道下水道汇入这套系统，整项基础设施用了3.18亿块砖。他算出各条管道所需的内径后，又将直径加倍。系统主干埋在当时新建的维多利亚、阿尔伯特和切尔西河堤之下。管网到1875年基本建成，大部分至今仍在使用。

然而，这项公共卫生工程建立在一个错误观念之上。当时占主导的“瘴气说”把疾病归因于污浊空气与蒸气，因此修下水道主要是为了压住臭味。它们却因另一个并非设计初衷的原因而奏效：保护了伦敦人的饮用水水质。

1866年霍乱再次出现，在伦敦东区夺走约5600条生命；那里正是唯一尚未接入管网的地区。

原本冲着鼻子去的补救措施，保护的却是肠胃。

核心词汇

先记这22项。连着下方的原文短语一起回想。

tolerable adj.
可忍受的
to make the air tolerable

antiquated adj.
陈旧过时的
Antiquated sewers and outlets

discharge v.
排放
discharged the untreated sewage

untreated adj.
未经处理的
the untreated sewage

sewage n.
污水
the untreated sewage

denounce v.
公开斥责
denounced the river

bill n.
法案
A bill funding new sewers

fast-track v.
加速办理；加快审议
fast-tracked into law

mastermind v.
主持策划
masterminded the response

intercepting sewer 名词短语
截流下水道
stout intercepting sewers

divert v.
使改道
diverted it east

infrastructure n.
基础设施
the infrastructure took

adequate adj.
足够的
an adequate calibre

calibre n.
管道内径
an adequate calibre for each pipe

sanitation n.
公共卫生设施
that sanitation rested on a fallacy

fallacy n.
错误观念；谬误
rested on a fallacy

predominant adj.
占主导的
The predominant belief

suppress v.
压制；消除
to suppress the smell

unintended adj.
非本意的；未预期的
an unintended reason

purity n.
洁净
the purity of the water

resurface v.
再次出现
Cholera resurfaced in 1866

aim at 动词短语
针对；旨在改善
A remedy aimed at the nose

文中词汇与搭配

补充词汇共42项。保留每个表达在本文中的意思。

fierce heat 名词短语
酷热

Houses of Parliament 专名
英国议会大厦

lime n.
石灰；文中为含氯漂白剂成分

bleach n.
漂白剂

fester v.
腐败发臭

contemplate v.
考虑

lavatory n.
厕所

outlet n.
排污口

dweller n.
居民

current n.
水流

excrement n.
粪便

stew v.
在闷热中沤积

decay v.
腐败；腐烂

vile adj.
恶心的；恶劣的

fume n.
难闻的气体

overpowering adj.
强烈得难以忍受的

episode n.
事件；一段经历

stink n.
恶臭

disgust n.
厌恶

Chancellor of the Exchequer 专名
英国财政大臣

as prep.
是；引出斥责中的定性

文中词汇与搭配 续

补充词汇共42项。保留每个表达在本文中的意思。

hellish adj.
地狱般的

unbearable adj.
难以忍受的

stout adj.
坚固的

short of 介词短语
在到达……之前

downstream adv.
向下游

tributary n.
支流；汇入主干的支管

brick n.
砖块

double v.
使加倍

backbone n.
主干

bury v.
埋在地下

underneath prep.
在……下面

newly adv.
新近地

substantially adv.
基本上；大体上

in use 介词短语
在使用中

rest on 动词短语
建立在……之上

belief n.
观念；看法

foul air 名词短语
污浊空气

vapour n.
蒸气

principally adv.
主要地

locality n.
地区

gut n.
肠胃

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The engineer Joseph Bazalgette _____ the response.

02 Having calculated an adequate calibre for each pipe, he _____ it.

03 Yet that sanitation rested on a _____.

04 A remedy _____ the nose protected the gut.

文章主线

- 危机 1858年酷暑，伦敦未经处理的污水在泰晤士河发臭，恶臭逼得议员考虑搬离议会。
- 行动 财政大臣公开抨击后，为新下水道拨款的法案约18天获通过，工程师巴泽尔杰特主持建设。
- 工程 82英里截流管拦下原本入河的污水，改送城市下游；按需要算出的管径再加倍。
- 误判 修下水道本为消除人们误认为会致病的臭气；实际作用是减少饮用水遭污水污染。
- 对照 1866年，唯一尚未接网的东区有约5600人死于霍乱；管网1875年基本完工，多数设施至今仍用。

参考答案　1 masterminded 2 doubled 3 fallacy 4 aimed at

原子弹之父的审判

The Trial of the Father of the Atomic Bomb



造出原子弹的科学家，为何被撤销涉密资格？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

J. Robert Oppenheimer, born in New York on 22 April 1904, was an accomplished physicist and prominent lecturer at Berkeley and Caltech. From April 1943 he oversaw a secretive weapons installation at Los Alamos, whose bombs were the collaborative creation of a thousands-strong workforce.

In the pre-dawn darkness of 16 July 1945, the first nuclear test illuminated the desolate New Mexico landscape with a blinding glare, its explosive force roughly 20 kilotons. On 6 and 9 August 1945 atomic bombings devastated Hiroshima and Nagasaki, where year-end fatalities numbered well over 100,000 and about 70,000 respectively.

Oppenheimer left Los Alamos in October 1945 and from 1947 presided over Princeton's Institute for Advanced Study. In October 1949 the advisory committee he chaired, expressing misgivings, unanimously counselled against a crash programme for the hydrogen bomb, a vastly more destructive weapon. In December 1953 a presidential directive severed his access to confidential documentation.

From 12 April to 6 May 1954 he was cross-examined at a security hearing, an administrative proceeding, not a criminal prosecution. On 29 June 1954 the Atomic Energy Commission voted 4 to 1 against letting him regain his clearance, two days before his consultancy contract was due to expire. The ruling did not find him disloyal, but nonetheless questioned his character and associates.

On 2 December 1963 President Johnson bestowed on him the Fermi Award, which Kennedy had authorised before his assassination. Oppenheimer died on 18 February 1967. In December 2022 the energy secretary officially vacated the 1954 determination, calling the process flawed and contrary to the commission's rules. The stigma of the revocation had persisted for fifty-five years beyond his lifetime.

中文译文

J.罗伯特·奥本海默于1904年4月22日生于纽约。他是卓有成就的物理学家，也是伯克利和加州理工学院的知名教师。从1943年4月起，他主管洛斯阿拉莫斯的一处秘密武器设施，那里的原子弹是数千名工作人员协作的成果。

1945年7月16日黎明前，首次核试验的刺眼强光照亮了新墨西哥州荒凉的地貌，爆炸威力约为2万吨TNT当量。1945年8月6日和9日，原子弹分别重创广岛、长崎；截至当年年底，两地死亡人数分别远超10万和约7万。

奥本海默于1945年10月离开洛斯阿拉莫斯，从1947年起执掌位于普林斯顿的高等研究院。1949年10月，他主持的顾问委员会表达疑虑，一致建议反对一项加速研制氢弹的紧急计划；这种武器的破坏力要大得多。1953年12月，一道总统指令切断了他接触机密文件的权限。

1954年4月12日至5月6日，他在安全听证会上接受盘问。这是行政程序，而非刑事起诉。6月29日，原子能委员会以4比1投票反对恢复他的涉密资格，距离他的顾问合同到期仅剩两天。裁决没有认定他不忠，却仍然质疑他的品格和交往对象。

1963年12月2日，约翰逊总统向他颁授费米奖；肯尼迪遇刺前已经批准授奖。奥本海默于1967年2月18日去世。2022年12月，能源部长正式撤销1954年的裁定，称当年的程序存在缺陷，违反了委员会的规定。撤销涉密资格所带来的污名，在他去世后又延续了55年。

核心词汇

先记这22项。连着下方的原文短语一起回想。

accomplished adj.
卓有成就的
an accomplished physicist

prominent adj.
知名的；杰出的
prominent lecturer

oversee v.
主管；监督
he oversaw

collaborative adj.
共同协作的
the collaborative creation

workforce n.
全体工作人员；劳动力
a thousands-strong workforce

illuminate v.
照亮
the first nuclear test illuminated

devastate v.
重创；摧毁
devastated Hiroshima and Nagasaki

fatality n.
死亡人数；死者
year-end fatalities

preside over 动词短语
主持；掌管
presided over Princeton's Institute

misgiving n.
疑虑；担忧
expressing misgivings

unanimously adv.
一致地
unanimously counselled against

counsel against 动词短语
建议不要；劝阻
counselled against a crash programme

directive n.
指令
a presidential directive

confidential adj.
机密的
confidential documentation

hearing n.
听证会
a security hearing

administrative adj.
行政的
an administrative proceeding

prosecution n.
刑事起诉
not a criminal prosecution

clearance n.
接触机密的许可
regain his clearance

expire v.
到期
was due to expire

vacate v.
撤销裁定
vacated the 1954 determination

flawed adj.
有缺陷的
calling the process flawed

stigma n.
污名
The stigma of the revocation

文中词汇与搭配

补充词汇共42项。保留每个表达在本文中的意思。

lecturer n.
授课教师

secretive adj.
秘密的；隐秘的

installation n.
大型专用设施

bomb n.
炸弹

creation n.
创造成果

thousands-strong adj.
有数千人的

pre-dawn adj.
黎明前的

desolate adj.
荒凉的

landscape n.
地貌；景观

blinding glare 名词短语
刺眼的强光

explosive adj.
爆炸的

bombing n.
轰炸

year-end adj.
截至年底的；年末的

respectively adv.
分别

Oppenheimer 专有名词
奥本海默

advisory adj.
顾问的；咨询的

crash programme 名词短语
紧急加速计划

hydrogen n.
氢

vastly adv.
大大地

destructive adj.
破坏性强的

presidential adj.
总统的

文中词汇与搭配 续

补充词汇共42项。保留每个表达在本文中的意思。

sever v.
切断

access to 名词短语
接触或使用……的权限

documentation n.
文件；书面资料

cross-examine v.
交叉盘问；详细询问

proceeding n.
程序；审理过程

atomic energy 名词短语
原子能

regain v.
重新获得

consultancy n.
顾问工作

disloyal adj.
不忠的

nonetheless adv.
尽管如此；仍然

associate n.
交往的人；同事

bestow on 动词短语
授予

authorise v.
批准；授权

assassination n.
暗杀；遇刺

secretary n.
此处指部长

officially adv.
正式地

determination n.
裁定；正式决定

contrary to 介词短语
违反；与……相反

revocation n.
撤销许可等

persist v.
持续存在

lifetime n.
一生

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the first nuclear test _____ the desolate New Mexico landscape

02 expressing _____,

03 an _____ proceeding, not a criminal prosecution.

04 calling the process _____ and contrary to the commission's rules.

文章主线

- 造弹 物理学家奥本海默领导数千人造出原子弹，1945年完成首次核试验，随后原子弹用于轰炸广岛、长崎。
- 分歧 战后，他主持的顾问委员会一致反对紧急推进氢弹计划；氢弹是一种比原子弹破坏力更强的核武器。
- 听证 1954年他接受安全听证，这是行政审查，并非刑事起诉；裁定也未认定他不忠。
- 失权 原子能委员会以4比1拒绝恢复他接触机密资料的资格，质疑其品格和交往，使其背负污名。
- 撤销 1963年虽获费米奖，1954年裁定直到2022年才由能源部长以程序违规为由撤销，距他去世已55年。

参考答案　1 illuminated 2 misgivings 3 administrative 4 flawed

一颗刺果钩出的发明

The Burrs That Became Velcro



衣服上的刺果，怎样变成了日常搭扣？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In 1941 George de Mestral, a Swiss engineer, returned from hunting in the Alps with his hound, an Irish Pointer. Prickly burdock seed-heads had hitched themselves to his clothes and lay tangled in the dog's fur. Under a microscope he saw thousands of tiny barbs, each a hook grasping the small loops in cloth.

Few took his idea seriously at first. In Lyon, then a weaving centre, a cotton version proved too frail to survive abrasion. Working with a textiles firm, he turned to nylon, whose hooks were stiff enough to hold yet easy to pull apart. The trial-and-error at the looms and mills lasted about ten years.

The name Velcro merged two French words, velours, meaning velvet, and crochet, meaning hook. A US patent duly followed in 1955. In 1959 a New York fashion show sent apparel stitched with the fasteners down the runway. The public was less charmed than expected, mostly because of the aesthetic.

The underdog proved its utility in outer space. NASA used it in the Apollo programme to keep gear anchored in weightlessness, so that nothing would hover in mid-air. Apollo 11 was equipped with a sizeable amount, about 3,300 square inches. NASA did not invent the nylon clasp.

When the patent ran out in 1978, cut-price knock-offs from other countries overran the market, and a niche product went mainstream. In 2017, to stop its brand-name melting into a generic noun, the company asked people to call such mimics hook and loop fasteners. In effect, the firm was asking the public to stop using its own name for rivals' goods.

中文译文

1941年，瑞士工程师乔治·德·梅斯特拉尔带着自己的猎犬——一只爱尔兰指示犬——从阿尔卑斯山狩猎归来。带刺的牛蒡种子头钩在衣服上，也缠进狗毛里。他用显微镜观察，发现上面有成千上万根微小倒刺，每根都像钩子，扣住织物上的小毛圈。

起初，几乎没人认真对待他的想法。他在当时的纺织中心里昂试制棉制搭扣，但它太脆弱，经不住磨损。与一家纺织公司合作时，他改用尼龙：这种材料的钩子足够挺硬，既能扣牢，又容易拉开。织机旁、工厂里的反复试验持续了约十年。

Velcro这个名字合并了两个法语词：velours意为天鹅绒，crochet意为钩子。1955年，美国专利随之获批。1959年，纽约一场时装秀让缝有这类搭扣的服装登上T台。然而公众的反应不如预期热烈，主要原因是外观。

这件起初不被看好的发明，在外太空证明了用途。美国航天局在阿波罗计划中用它把装备固定住，避免装备在失重环境里漂浮。阿波罗11号配备的搭扣面积相当可观，约3300平方英寸。但这种尼龙扣件并非美国航天局发明。

1978年专利到期后，其他国家的廉价仿品涌入市场，小众产品也走向大众。2017年，为防品牌名逐渐变成整类产品的通用名词，公司要求人们把这类仿品称为“钩毛搭扣”。实际上，它是在请公众不要再用自家的名字称呼竞争对手的商品。

核心词汇

先记这22项。连着下方的原文短语一起回想。

barb n.
倒刺
thousands of tiny barbs

grasp v.
抓住；扣住
a hook grasping the small loops

loop n.
毛圈；环
the small loops in cloth

frail adj.
脆弱的
a cotton version proved too frail

abrasion n.
摩擦造成的磨损
to survive abrasion

nylon n.
尼龙
he turned to nylon

stiff adj.
挺硬的；不易弯曲的
hooks were stiff enough to hold

pull apart 动词短语
拉开；扯开
easy to pull apart

trial and error n.
反复试验；试错
The trial-and-error

merge v.
合并
merged two French words

apparel n.
服装
apparel stitched with the fasteners

fastener n.
扣件；搭扣
apparel stitched with the fasteners

aesthetic n.
外观；审美风格
mostly because of the aesthetic

utility n.
实用性；用途
proved its utility in outer space

anchor v.
固定住
keep gear anchored in weightlessness

hover v.
悬浮；漂浮
hover in mid-air

equip with 动词短语
配备
was equipped with a sizeable amount

run out 动词短语
期限届满
the patent ran out in 1978

niche adj.
小众的
a niche product went mainstream

mainstream adj.
主流的；大众化的
a niche product went mainstream

generic adj.
通用的；非特定品牌的
a generic noun

in effect 副词短语
实际上
In effect, the firm

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

Swiss adj.
瑞士的

hound n.
猎犬

prickly adj.
多刺的

seed-head n.
含种子的花序；种子头

hitch v.
钩住；挂住

tangle v.
缠结

fur n.
动物皮毛

hook n.
钩子

take v.
与seriously连用，认真对待

seriously adv.
认真地

weaving n.
织造；纺织

textile n.
纺织品

loom n.
织机

mill n.
工厂；纺织厂

velvet n.
天鹅绒

duly adv.
如期；按预期

stitch v.
缝上

runway n.
时装表演的T台

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

charm v.
使着迷；吸引

mostly adv.
主要地；大多

underdog n.
不被看好的一方

outer space 名词短语
外太空

Apollo programme 专有名词
阿波罗计划

gear n.
装备

in weightlessness 介词短语
在失重环境中

mid-air n.
半空中

sizeable adj.
相当大的；可观的

clasp n.
扣件

cut-price adj.
廉价的；降价的

knock-off n.
仿制品

overrun v.
大量涌入；占满

brand name n.
品牌名

melt into 动词短语
逐渐融为；变成

noun n.
名词

mimic n.
仿制品

rival n.
竞争对手

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Under a microscope he saw thousands of tiny _____,

02 he turned to _____, whose hooks were stiff enough to hold yet easy to pull apart.

03 The underdog proved its _____ in outer space.

04 a niche product went _____.

文章主线

- 启发 瑞士工程师梅斯特拉尔发现刺果用微钩扣住衣物毛圈，便仿照它设计可反复扣合、拉开的搭扣。
- 试制 棉制搭扣不耐磨，他改用尼龙，反复试验约十年，1955年获得美国专利。
- 应用 公众主要因外观而反应冷淡；美国航天局在阿波罗计划中用搭扣固定失重中的装备，并非搭扣的发明者。
- 普及 1978年专利到期后，廉价仿品涌入市场，原本小众的搭扣逐渐成为日常用品。
- 正名 Velcro是品牌名；2017年品牌方要求将仿品称为钩毛搭扣，避免自家商标泛指整类产品。

参考答案　1 barbs 2 nylon 3 utility 4 mainstream

破解霍乱之谜的医生

The Doctor Who Solved the Cholera Mystery



破解霍乱之谜，关键是地图还是饮水？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In the first ten days of September 1854, cholera ravaged an overcrowded corner of London's Soho; upwards of 500 people perished. Most doctors attributed epidemics to miasma, foul air arising from filth. John Snow, a physician who had been convinced since 1849 that cholera's poison was swallowed, not inhaled, suspected the Broad Street pump.

Finding little visible impurity, he hesitated, then consulted the death register and questioned bereaved families. Most victims lived near the pump and drank from it. At his urging, parish officials removed the handle, but daily fatal attacks had already plunged from a peak of 143 to 12. He later mapped the deaths.

In 1855 Edmund Parkes, a professor of clinical medicine, interpreted the cluster on that map as a poison drifting through the air. Yet a map could chart where people died, not what they drank.

The crucial evidence, gathered in 1854, lay in exceptions. Amid the deaths in the surrounding streets, a workhouse with its own well lost a mere five of 535 inmates. More than 70 nearby brewery workers, with a beer allowance and their own well, escaped severe cholera.

Conversely, a Hampstead widow who had not set foot in Soho for months had the pump's water carted to her daily. She died, as did a visiting niece who drank it. Hampstead then had no cholera.

In 1855 a government inquiry declined to condemn the pump, contending its water had at most absorbed airborne poison. Meanwhile, the sceptical curate Henry Whitehead set out to refute Snow but found that those who drank from the pump were far more likely to succumb to the disease than others, vindicating him.

中文译文

1854年9月的头十天，霍乱在伦敦苏豪区一片人口过密的街区肆虐，500多人丧生。多数医生把流行病归因于瘴气，也就是污秽中产生的恶臭空气。医生约翰·斯诺自1849年起便确信，霍乱的致病物是被吞入人体，而不是随空气吸入；他因此怀疑宽街的公共水泵。

泵水中几乎没有肉眼可见的杂质，这让他一度犹豫。随后，他查阅死亡登记册，走访失去亲人的家庭。多数死者住在水泵附近，也喝过那里的水。在他的敦促下，教区官员拆下了泵柄；但在此之前，每天新发的致死病例数，就已经从高峰的143例骤降到12例。他后来才把死亡地点标在地图上。

1855年，临床医学教授埃德蒙·帕克斯把地图上的死亡聚集现象，解释成毒物在空气中飘散的结果。然而，地图只能记录人们死在哪里，并不能说明他们喝过什么。

真正关键的证据早在1854年就已收集到，存在于那些例外之中。周边街巷死亡接连发生，一所自备水井的济贫院，535名被收容者中却只有5人死亡。附近啤酒厂的70多名工人，既享有啤酒配给，也有自备水井，没有患上重症霍乱。

与之相反，汉普斯特德的一位寡妇，已经好几个月没有踏入苏豪区，却每天让人把宽街水泵的水用车运到家中。她去世了；来探望她、也喝了这些水的侄甥女，同样去世了。而汉普斯特德当时并没有流行霍乱。

1855年，政府的一次调查拒绝认定水泵应当负责，认为泵水充其量只是吸收了空气中的毒物。与此同时，原本持怀疑态度的助理牧师亨利·怀特黑德，着手寻找证据来驳倒斯诺，却发现喝过泵水的人，死于这种疾病的可能性远高于其他人，反倒印证了斯诺的判断。

niece可指侄女或外甥女，原文未进一步区分，译文以“侄甥女”统称。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ravage v.
肆虐；严重毁坏
cholera ravaged an overcrowded corner

attribute v.
归因于
doctors attributed epidemics

foul adj.
恶臭的；污浊的
foul air arising from filth.

physician n.
医生
John Snow, a physician

convince v.
使确信；此处指深信不疑
convinced since 1849

consult v.
查阅
then consulted the death register

bereaved adj.
失去亲人的
questioned bereaved families.

fatal adj.
致死的
daily fatal attacks

plunge v.
骤降
had already plunged

interpret v.
解释；解读
interpreted the cluster

cluster n.
聚集；密集的一群
the cluster on that map

crucial adj.
至关重要的
The crucial evidence

exception n.
例外
lay in exceptions.

conversely adv.
相反地；反过来
Conversely, a Hampstead widow

inquiry n.
调查；官方调查
a government inquiry

decline to phr. v.
拒绝做……
declined to condemn the pump

contend v.
主张；坚持认为
contending its water had

airborne adj.
经空气传播的
airborne poison.

sceptical adj.
持怀疑态度的
the sceptical curate Henry Whitehead

refute v.
驳倒；反驳
set out to refute Snow

succumb to phr. v.
死于（疾病）；屈服于
succumb to the disease

vindicate v.
证明……正确；印证
vindicating him.

文中词汇与搭配

补充词汇共35项。保留每个表达在本文中的意思。

overcrowded adj.
人口过密的；过度拥挤的

upwards of phr.
超过

perish v.
丧生

epidemic n.
流行病；疫情

to prep.
与attribute连用，表示归因于

arise v.
产生；源于

filth n.
污秽

swallow v.
吞入

inhale v.
吸入

hesitate v.
犹豫

register n.
登记册

victim n.
受害者；此处指死者

urging n.
敦促；催促

parish n.
教区

peak n.
峰值；最高点

clinical medicine n.
临床医学

poison n.
毒物；毒素

drift v.
飘散

文中词汇与搭配 续

补充词汇共35项。保留每个表达在本文中的意思。

chart v.
绘制；记录分布

lie in phr. v.
在于；存在于

amid prep.
在……之中

surrounding adj.
周围的

workhouse n.
济贫院

a mere phr.
仅仅；区区

inmate n.
被收容者

allowance n.
配给；补助

widow n.
寡妇

set foot in phr.
踏入

cart v.
用车运送

niece n.
侄女；外甥女

then adv.
当时

condemn v.
谴责；认定应负责任

absorb v.
吸收

meanwhile adv.
与此同时

set out to phr. v.
着手去；打算去

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 At his urging, parish officials removed the handle, but daily _____ attacks had already plunged from a peak of 143 to 12.

02 In 1855 Edmund Parkes, a professor of clinical medicine, interpreted the _____ on that map as a poison drifting through the air.

03 The _____ evidence, gathered in 1854, lay in exceptions.

04 _____, a Hampstead widow who had not set foot in Soho for months had the pump's water carted to her daily.

文章主线

- 旧说 1854年伦敦苏豪区霍乱十日夺命逾500人；多数医生归因臭气，斯诺则怀疑饮水传播。
- 追查 他查死亡记录、访问家属，发现许多死者喝过宽街水泵的水，促使官员拆下泵柄。
- 对照 自备井的济贫院仅少数死亡，啤酒厂无人患重症；远处无疫情地区，却有两人喝泵水后死亡。
- 辨别 每日致死病例数在拆泵柄前已大幅减少；地图只标死亡地点，关键是比较人们是否喝过泵水。
- 验证 官方调查仍坚持空气传播；原想驳倒斯诺的牧师怀特黑德却发现喝泵水者更易死亡，支持了斯诺。

参考答案　1 fatal 2 cluster 3 crucial 4 Conversely

拥挤的苍穹

The Crowded Heavens



两颗卫星相撞，为什么多年后仍要避让？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 10 February 2009, nearly 790 kilometres above Siberia, two satellites slammed together at about 42,000 kilometres an hour. Iridium 33 belonged to an American satellite-phone constellation; Kosmos 2251, a Russian military satellite, had been derelict since about 1995. Both shattered in the first accidental collision between two intact satellites.

Neither craft swerved. The Russian hulk, lacking propulsion, could not. A public collision-warning service had forecast a narrow miss of about 584 metres, yet the predicted encounter was not among the week's closest; such warnings were frequent and imprecise. One owner's discarded relic demolished another's working asset.

The wreckage, however, endangered everyone else. The impact flung out more than 2,000 catalogued fragments and a multitude of others too small to pinpoint. At that height air drag is feeble, so many would linger for decades. In January 2012, the International Space Station fired its thrusters to dodge an Iridium 33 remnant.

In June 1978, two NASA scientists envisaged a vicious spiral, later named the Kessler syndrome: each collision would scatter fragments, raising the odds of subsequent crashes. The 2009 crash illustrated that premise without triggering a runaway chain reaction. By the end of 2024, surveillance networks tracked about 40,000 objects, only some 11,000 of them functional.

Such a cascade would unfold over decades, though, and is not inevitable; the risk can be mitigated. Operators now routinely manoeuvre satellites out of harm's way. In September 2022, America's Federal Communications Commission, which licenses satellites, mandated disposal of low-orbit craft within five years of retirement instead of 25. Years after the crash, spacecraft whose owners had no part in it kept swerving around its splinters.

中文译文

2009年2月10日，西伯利亚上空近790公里处，两颗卫星以约每小时4.2万公里的相对速度猛撞在一起。铱星33号属于美国的一个卫星电话星座；俄罗斯军用卫星宇宙2251号则约从1995年起便已失效。两星双双解体，这是两颗完整卫星之间首次发生意外相撞。

两颗卫星都没有转向避让。俄方那颗失效卫星缺乏推进能力，无法躲开。一个公开的碰撞预警服务曾预测，两星会相距约584米擦过，但这次预计交会还不在当周距离最近的几次之列；这样的预警既频繁，又不精确。一方丢弃的旧物，摧毁了另一方仍在工作的资产。

碎片却危及所有其他航天器。撞击抛出了两千多块已编目碎片，还有大量小得无法精确定位的碎片。那个高度空气阻力很弱，许多碎片会在轨道上滞留数十年。2012年1月，国际空间站启动推进器，避开了一块铱星33号的残片。

1978年6月，两位美国航天局科学家设想了一种后来被称为“凯斯勒综合征”的恶性循环：每次碰撞都会散出碎片，提高后续相撞的概率。2009年的事故展示了这一机制的基础，却没有引发失控的连锁反应。截至2024年底，监测网络跟踪着约4万个在轨物体，其中只有约1.1万个正常运行。

不过，这样的连锁过程会历时数十年，并非不可避免，风险也能减轻。卫星运营方如今会经常调整卫星轨道，避开危险。2022年9月，负责颁发卫星许可的美国联邦通信委员会要求，其监管的低轨航天器在退役后五年内完成处置，而不再是25年。事故多年后，那些所有者与此事毫无关系的航天器，仍要一再转向躲避当年的碎片。

核心词汇

先记这22项。连着下方的原文短语一起回想。

derelict adj.
已废弃失效的
had been derelict since about 1995

collision n.
碰撞
the first accidental collision

propulsion n.
推进力；推进能力
lacking propulsion

narrow miss 名词短语
险些相撞；近距离擦过
a narrow miss of about 584 metres

imprecise adj.
不精确的
frequent and imprecise

discard v.
弃置
One owner's discarded relic

asset n.
资产
another's working asset

wreckage n.
残骸；碎片
The wreckage, however

catalogue v.
编目；登记
more than 2,000 catalogued fragments

pinpoint v.
精确定位
too small to pinpoint

air drag 名词短语
空气阻力
air drag is feeble

linger v.
滞留；迟迟不消失
many would linger for decades

dodge v.
躲避
to dodge an Iridium 33 remnant

envisage v.
设想
scientists envisaged a vicious spiral

subsequent adj.
后续的
the odds of subsequent crashes

premise n.
前提；基本设想
illustrated that premise

trigger v.
引发
without triggering

chain reaction 名词短语
连锁反应
a runaway chain reaction

surveillance n.
监视；监测
surveillance networks tracked

functional adj.
正常运行的
11,000 of them functional

inevitable adj.
不可避免的
is not inevitable

disposal n.
处置
mandated disposal of low-orbit craft

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

slam v.
猛撞

belong to 动词短语
属于

constellation n.
卫星星座；卫星群

shatter v.
粉碎；解体

accidental adj.
意外的

swerve v.
突然转向；转向避让

hulk n.
废弃的大型躯壳

encounter n.
交会；相遇

frequent adj.
频繁的

relic n.
遗留旧物

demolish v.
摧毁

endanger v.
危及

impact n.
撞击

fling out 动词短语
抛出；甩出

a multitude of 名词短语
大量的

feeble adj.
微弱的

space station 名词短语
空间站

fire v.
启动；点火

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

thruster n.
推进器

remnant n.
残片；遗留部分

vicious spiral 名词短语
恶性循环

syndrome n.
综合征

scatter v.
散布；使散开

raise the odds of 动词短语
提高……发生的概率

illustrate v.
例证；说明

runaway adj.
失控的

unfold v.
逐步展开

mitigate v.
减轻；缓解

routinely adv.
例行地；经常地

manoeuvre v.
操纵；调整运行轨迹

out of harm's way 介词短语
避开危险；在安全处

mandate v.
强制要求

retirement n.
退役

instead of 介词短语
而不是；代替

have no part in 动词短语
与……无关；未参与

splinter n.
碎片

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the first accidental _____ between two intact satellites.

02 The Russian hulk, lacking _____, could not.

03 a multitude of others too small to _____.

04 and is not _____; the risk can be mitigated.

文章主线

- 相撞 2009年，两颗完整卫星首次意外相撞：失效的宇宙2251号撞上仍工作的铱星33号。
- 预警 公开预警频繁且不准，预计两星会隔约584米擦过，并非当周最近交会；失效卫星也无法避让。
- 现状 碰撞留下两千多块已编目碎片；截至2024年底，全球跟踪约4万个在轨物体，只有约1.1万个正常运行。
- 风险 高空阻力小，碎片可滞留数十年，又增加碰撞风险；此次未触发失控连锁反应，恶性循环并非必然。
- 应对 卫星运营方加强避让；美国联邦通信委员会把其监管低轨卫星的退役后处置期限，从25年缩至5年。

参考答案　1 collision 2 propulsion 3 pinpoint 4 inevitable

被降级的冥王星

The Planet That Was Demoted



冥王星本身没变，为什么会失去行星身份？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 18 February 1930, comparing photographic plates at the Lowell Observatory, the astronomer Clyde Tombaugh perceived a dim dot that shifted between them. An eleven-year-old English girl put forward the name, after the Roman deity of the underworld. For 76 years Pluto held the ninth slot on the planetary roster.

In January 2005, astronomers found a far-off body, later named Eris, at first thought larger than Pluto. NASA briefly called it the tenth planet. The contender posed an awkward dilemma. The term "planet" had hitherto been used loosely, and with more icy worlds likely to be found, a precise definition became a necessity.

On 24 August 2006 the International Astronomical Union, convened in Prague, voted that a planet must revolve around the Sun, be nearly spherical, and have cleared the neighbourhood around its orbit. Pluto failed that last criterion and was reclassified as a dwarf planet, leaving eight. Pluto was physically unaltered; only the boundary had moved.

The vote was valid, yet its legitimacy drew criticism: only 424 of some 10,000 members took part. Hundreds of dissenting astronomers signed a petition. The American Dialect Society made "plutoed", playful slang for "to devalue", its word of 2006, and New Mexico and Illinois lawmakers passed resolutions affirming that Pluto was a planet.

Amid the controversy, an irony was already in transit. New Horizons was launched on 19 January 2006, seven months before the vote, carrying a pinch of Tombaugh's ashes. On 14 July 2015 the unmanned craft flew past Pluto, transmitting images of an enormous heart-shaped plain, fittingly named Tombaugh Regio. The discoverer's ashes swept past the heart that bears his name.

中文译文

1930年2月18日，天文学家克莱德·汤博在洛厄尔天文台比较照相底片时，发现一个暗淡的小点在不同底片之间移动。一名11岁的英国女孩建议，以罗马神话中的冥界之神为它命名。此后76年，冥王星一直占据行星名单上的第九个位置。

2005年1月，天文学家发现一个遥远天体，后来命名为厄里斯，起初认为它比冥王星更大。NASA曾短暂地称它为第十大行星。这个新候选者带来一道棘手难题：“行星”此前一直是个使用宽泛的称呼，而随着更多冰冷天体可能被发现，给它下一个精确定义成了必要之举。

2006年8月24日，国际天文学联合会在布拉格开会，投票决定：行星必须绕太阳运行，形状近似球体，并已在轨道邻域占据引力主导地位。冥王星不符合最后一条标准，被重新归为矮行星，太阳系行星由此只剩八颗。冥王星本身丝毫未变，变化的只是分类界线。

表决有效，但其正当性遭到批评：约一万名会员中，只有424人参与。数百名持异议的天文学家联名请愿。美国方言学会把戏谑地表示“贬低”的俚语“plutoed”选为2006年度词汇；新墨西哥州和伊利诺伊州的立法者则通过决议，确认冥王星是行星。

争论之中，一件颇具反讽意味的事早已在路上。新视野号于2006年1月19日发射，比那场表决早七个月，携带着一小撮汤博的骨灰。2015年7月14日，无人探测器飞掠冥王星，传回巨大心形区域的图像；这个区域恰以汤博的名字命名为“汤博区”。发现者的骨灰，掠过了那颗以他命名的“心”。

“清理轨道邻域”指在该区域占据引力主导地位，并非清空所有天体。英文plain是对心形地貌的概括；译文用“心形区域”，因为整个汤博区并不全是平原。

核心词汇

先记这22项。连着下方的原文短语一起回想。

perceive v.
察觉；注意到
perceived a dim dot

put forward 动词短语
提出；提议
put forward the name

contender n.
竞争者；候选者
The contender posed an awkward dilemma

dilemma n.
两难困境；棘手难题
an awkward dilemma

precise adj.
精确的
a precise definition

definition n.
定义
a precise definition

necessity n.
必要之事
became a necessity

convene v.
开会；召开会议
convened in Prague

revolve around 动词短语
绕……运行
revolve around the Sun

spherical adj.
球形的
be nearly spherical

criterion n.
标准
that last criterion

reclassify as 动词短语
重新归类为
reclassified as a dwarf planet

unaltered adj.
未改变的
Pluto was physically unaltered

boundary n.
界线；此处指分类边界
only the boundary had moved

valid adj.
有效的
The vote was valid

legitimacy n.
正当性
its legitimacy drew criticism

dissent v.（分词作定语）
持异议
dissenting astronomers

petition n.
请愿书
signed a petition

affirm v.
确认；明确主张
affirming that Pluto was a planet

controversy n.
争议
Amid the controversy

irony n.
具有反讽意味的事
an irony was already in transit

transmit v.
传送；传回
transmitting images

文中词汇与搭配

补充词汇共31项。保留每个表达在本文中的意思。

photographic adj.
照相的

astronomer n.
天文学家

dim adj.
暗淡的

dot n.
小点

deity n.
神灵

underworld n.
冥界

slot n.
位置；名额

planetary adj.
行星的

roster n.
名单

far-off adj.
遥远的

awkward adj.
棘手的；尴尬的

hitherto adv.
此前；迄今为止

loosely adv.
宽泛地；不严格地

icy adj.
冰冷的；覆冰的

neighbourhood n.
周边区域；此处指轨道邻域

dwarf adj.
矮的；用于矮行星名称

physically adv.
在实体上；在物理上

criticism n.
批评

playful adj.
戏谑的

slang n.
俚语

devalue v.
贬低；使贬值

lawmaker n.
立法者

resolution n.
决议

in transit 介词短语
在途中

a pinch of 数量短语
一小撮

unmanned adj.
无人的；无人驾驶的

enormous adj.
巨大的

heart-shaped adj.
心形的

fittingly adv.
恰如其分地

discoverer n.
发现者

sweep past 动词短语
飞掠而过

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The contender posed an awkward _____.

02 Pluto failed that last _____ and was reclassified as a dwarf planet, leaving eight.

03 Pluto was physically _____; only the boundary had moved.

04 The vote was valid, yet its _____ drew criticism: only 424 of some 10,000 members took part.

文章主线

- 发现 1930年汤博发现冥王星，此后76年，它一直被列为太阳系的第九颗行星。
- 重审 2005年发现的厄里斯起初被认为比冥王星更大，预期还有更多同类天体，行星需要明确标准。
- 改类 2006年新规要求行星绕日、近球形并靠引力主导轨道邻域；冥王星未过第三条，改列矮行星，行星剩八颗。
- 争议 国际天文学联合会的表决有效，但约一万会员仅424人参与，正当性受质疑；冥王星本身未变。
- 探访 无人探测器新视野号携少量汤博骨灰，于2015年飞掠冥王星，传回以这位发现者命名的心形区域图像。

参考答案　1 dilemma 2 criterion 3 unaltered 4 legitimacy